Khwaja Yunus Ali University

School of Science and Engineering

Dept. of Computer Science and Engineering

Mid-term Examination: Fall 2018

Program: B.Sc. in CSE (Batch: 6^{th}) 1^{st} Year 2^{nd} Semester

Course Code: PHY 1201

Course Title: Physics – II

Time: 1 hour & 30 minutes — Total Marks: 30

Group-B

[Answer any 5 (Five) questions including No.1] **5 × 5 = 25**

1. Read the stem given below and answer the following questions:

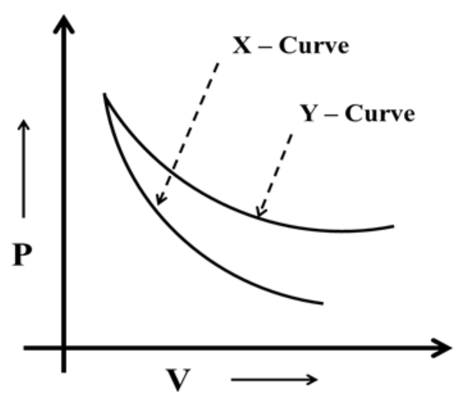
   Most of the processes in nature are x-processes. The entropy increases day by day in x-process. Whose end result is heat death of universe.

   a) What is entropy? 1

   b) Give the name of the x-process mentioned in the above stem. Also give an example of it. 1

   c) Justify the 2^{nd} sentence of above stem. 3

2. See the figure for thermodynamic system and answer the questions that follows:



   a) Define X & Y which are mentioned in above stem. 1+1=2

   b) Establish the relationship between slopes of two curves. 3

3. a) Draw the block diagram of heat engine. 2

   b) A Carnot's engine has 50% efficiency when the sink temperature is 27^{0}C. What is the increase in temperature of the heat source when its efficiency is 60%? 3

4. a) State Mayer's Hypothesis. 1

b) Write down the Van der Waal's Equation of State. 1

c) The volume of air is made three times the original volume by adiabatic expansion. If the initial pressure is 1 atmospheric pressure, what is the final pressure? [$\gamma = 1.40$] 3

5. a) What are the factors on which the mean free path depends. 2

b) The mean free path of the molecules of a gas is 6×10^{-8}m and the diameter of a molecule is 2.5×10^{-10}m. Find the number of molecules per m^3. 3

6. a) Explain the principle of temperature measurement. 2

b) A platinum resistance thermometer reads 2Ω and 2.73Ω at ice point and steam point respectively. Find the temperature at which it reads 4.83Ω. 3

7. a) What is Helmholtz free energy? 2

b) Define enthalpy. 1

c) Establish the relation between Helmholtz & Gibbs free energy. 2

Khwaja Yunus Ali University

School of Science and Engineering

Dept. of Computer Science and Engineering

Mid-term Examination: Fall 2018

Program: B.Sc. in CSE (Batch: 6^{th}) 1^{st} Year 2^{nd} Semester

Course Code: PHY 1201

Course Title: Physics – II

Time: 1 hour & 30 minutes | Total Marks: 30

Group-A

Time: 10 min | **[Answer all questions]** | **0.5 × 10 = 5**

Student Name:	**ID No.:**

1.	How many degrees of freedom in a rotational body?			
	a)	2	b)	3
	c)	4	d)	5
	e)	6		
2.	‘Production of heat by friction’ is the example of ……………………… Process.			
	a)	Reversible	b)	irreversible
	c)	adiabatic	d)	Isothermal
	e)	Isobaric		
3.	Platinum resistance thermometer can be used to measure the temperature from –			
	a)	-273^0C to 0^0C	b)	-273^0C to 273^0C
	c)	-273^0C to 600^0C	d)	-273^0C to 2000^0C
	e)	-250^0C to 1300^0C		
4.	……………………….. follow the Bayle’s law.			
	a)	Isothermal Process	b)	Adiabatic Process
	c)	Isochoric Process	d)	Isobaric Process
	e)	Reversible Process		

5.	Which one is the correct relation between C_p & C_v?			
	a)	$C_p < C_v$	b)	$C_p = C_v$
	c)	$C_p > C_v$	d)	$C_p + C_v = R$
	e)	$C_v - C_p = R$		
6.	What is the value of γ for Helium gas?			
	a)	1.33	b)	1.40
	c)	1.67	d)	1.81
	e)	1.97		
7.	If $W \infty Q$, what will be proportionality constant?			
	a)	μ	b)	G
	c)	R	d)	K
	e)	J		
8.	Which is related with thermo electric thermometer?			
	a)	Thermocouple	b)	Bayle's law
	c)	Heat death of universe	d)	Elastic collision
	e)	Entropy		
9.	What is the unit of entropy?			
	a)	Jkg^{-1}	b)	$calJ^{-1}$
	c)	$Jcal^{-1}$	d)	JK^{-1}
	e)	KJ^{-1}		
10.	For adiabatic process, -			
	a)	$T = constant$	b)	$V = constant$
	c)	$S = constant$	d)	$P = constant$
	e)	$Q = changeable$		

Invigilator Signature

Examiner Signature